J Therm Anal Calorim (2016) 125:255–269
DOI 10.1007/s10973-016-5380-3

Thermodynamic description of the Ni–Yb and Ni–Eu binary systems

Saïd Kardellass[1] · Colette Servant[2] · Najim Selhaoui[1]

Received: 27 September 2015 / Accepted: 25 February 2016 / Published online: 5 April 2016
© Akadémiai Kiadó, Budapest, Hungary 2016

Abstract Thermodynamic descriptions of the Ni–Yb and Ni–Eu binary systems have been performed by means of the CALculation of PHAse Diagrams approach on the basis of the experimental data including the phase equilibria and thermodynamic properties. The Gibbs free energies of the liquid and the terminal solid solutions (fcc_A1, bcc_A2 and hcp_A3) are described by the subregular solution model with the Redlich–Kister formalism, and those of the intermetallic compounds αNiYb=NiYb_LT, βNiYb=-NiYb_HT, Ni_2Yb, Ni_3Yb, Ni_5Yb and $Ni_{17}Yb_2$ in the Ni–Yb system and NiEu, Ni_2Eu, Ni_5Eu and $Ni_{17}Eu_2$ in the Ni–Eu system were described as strict stoichiometric compounds. As a result, two self-consistent thermodynamic data sets for describing the Ni–Yb and Ni–Eu binary systems have been obtained, and the calculated results show a satisfactory agreement with the corresponding experimental data.

Keywords Ni–Yb system · Ni–Eu system · Thermodynamic assessment · CALPHAD method · Redlich–Kister formalism

✉ Colette Servant
coletteservant@orange.fr

Saïd Kardellass
saidkardellass@yahoo.fr

[1] Laboratoire de Thermodynamique et Energétique, LTE, Université Ibn-Zohr, B.P. 8106, Agadir, Morocco

[2] Laboratoire de Physicochimie de l'Etat Solide, ICMMO, Université Paris-Sud, 91405 Orsay Cedex, France

Introduction

The rare earth elements (REEs), exhibiting good combinations of magnetic, optical, electrical and thermal properties, are widely used in high-performance functional materials, such as permanent magnets, magnetic refrigerants and hydrogen storage materials [1–3]. The intermetallic compounds containing nickel (Ni) and rare earth elements (REEs) may be of special interest due to a wide range of applications in Ni-based super alloys and in magnetic, composite, amorphous, hydrogen storage and quasi crystalline materials [4–9].

Considering a systematic design and processing of the materials, the knowledge of phase diagrams is of great importance. Generally, they have been only determined by experimental approach that is costly, meticulous and time-consuming. Thus, the computer simulation and the CALPHAD (CALculation of PHAse Diagrams) method come in and play an important role to cut down on cost and time during development of materials [10]. The CALPHAD method is based on the description of the Gibbs energies of the different phases present in a system. These are evaluated by least-square fitting of model parameters in order to describe the available experimental data including the phase diagram and thermodynamic data of a given system as well as possible.

In the present paper, as a part of the thermodynamic investigation of Ni–RE-based alloys (RE = Sc [11] and Ho [12]) which is intended to give a better understanding of the constitutional properties needed for the development of these alloys, our scope is to carry out thermodynamic descriptions for the phase diagrams both of the Ni–Eu and the Ni–Yb systems by the CALPHAD approach. The two systems have been previously assessed by Berezutskii et al. [13] with a CALPHAD approach but without using the

Redlich–Kister formalism [14] to optimize separately the thermodynamic parameters of the solution phases as we did in the present work. Indeed, they considered a linear dependence between the enthalpies and entropies of the phases.

Review on literature data

The Ni–Yb system

Phase diagram data

Using X-ray diffraction methods, Buschow [15] firstly studied the Ni–Yb binary system. Later on, Palenzona and Cirafici [16] measured the phase diagram of the Ni–Yb system by using differential thermal analysis, X-ray diffractography and metallographic observations. In this later work, the existence of five intermediate compounds, namely NiYb, Ni_2Yb, Ni_3Yb, Ni_5Yb and $Ni_{17}Yb_2$, with the first in low-temperature (α) and high-temperature (β) forms were established, as well as the limited terminal solid solutions and the region of liquid immiscibility (which has been found for Yb-rich alloys), although its extent has not been determined other than at the monotectic temperature at 1025 °C where it extends from 54 to 91 at.% Yb. Alloys in this composition range were heated in Mo containers at 1500 °C and shaken followed by rapid cooling at room temperature by water quenching. All of the alloys exhibited two layers; the upper layer was a matrix of Yb with NiYb particles, and the lower layer was a matrix of NiYb with Yb particles dispersed in it. The layer separation is expected because of the large difference in density between Yb (6.97 g/cm^3) and NiYb (10.08 g/cm^3). This latter work has been the source of information to assess and draw Fig. 1 (dashed line) by Nash [17]. This evaluation reported that the measured melting point and the bcc_A2 ↔ fcc_A1 transformation temperature for elemental Yb by Palenzona and Cirafici [16] were 825 and 740 °C, whereas the accepted values by Gschneidner and Calderwood [18] for pure Yb are 819 and 795 °C; these differences indicated, as reported by Palenzona and Cirafici [16], the presence of impurities in their Yb starting material. Thus, this presents a problem in evaluating the diagram by Palenzona and Cirafici [16], as the eutectic reaction Liq ↔ (Yb) + NiYb was reported to occur at 790 °C slightly below the accepted bcc_A2 ↔ fcc_A1 transformation in Yb. If the liquid is described as an ideal solution and NiYb as a stoichiometric phase, the liquidus temperature at the eutectic composition (98 at.% Yb) determined by Palenzona and Cirafici [16] is 793.12 °C. Also, as reported by Nash [17], because there is no solubility of Ni in (Yb) measurable from the lattice parameter variation, the calculated value is also too low. Thus, if we assume that the form of the diagram of Palenzona and Cirafici [16] at the Yb-rich end is correct, then we can say little more than that the eutectic temperature lies between 819 and 795 °C [17]. For the purpose of drawing Fig. 1 (dashed line), the eutectic temperature has been taken by Nash [17] as 805 °C and the bcc_A2 (Yb) ↔ fcc_A1 (Yb) + NiYb eutectoid temperature as 794 °C. Also, Nash [17] reported that Palenzona and Cirafici [16] found that for alloys between 0 and 25 at.% Yb, a reaction with the Mo crucibles occurred, so an alternative procedure was used. The alloys were melted in alumina crucibles, heated to 1450 °C and homogenized at 1000 °C for 2 days. Portions of the alloys were transferred to the differential thermal analysis (DTA) unit, presumably in Mo crucibles, and the thermal effects noted on heating were used to construct the portion of the diagram from 0 to 25 at.% Yb. Although not mentioned by Palenzona and Cirafici [16], Nash [17] reported also that a reaction with the crucible probably prevented cycled heating and cooling runs from giving reproducible results. Some faint thermal effects were observed that were reported as not reproducible. The data in this composition range should therefore be viewed with some caution (particularly the liquidus data), because it is likely that Mo contaminated the alloys [17].

Buschow [15] prepared alloys by heating the starting materials in a sealed Mo container at 1200 °C for 5 min, followed by annealing the Mo containers in evacuated quartz tubes for 3 weeks at 700–800 °C. He reported no reaction between the alloys and the Mo container during this procedure. According to Nash [17], attempts to produce a single-phase Ni_7Yb_2 alloy (a phase commonly observed in Ni–RE systems) were unsuccessful. However, X-ray lines were observed that belonged to a hexagonal phase with a large *c*/*a* ratio. This result is similar to that observed in the Ni–Er system and was confirmed by Palenzona and Cirafici [16]. Buschow [15] suggested that the stability of the Ni_7RE_2 phase depends on the size of the RE element, becoming unstable when the RE element becomes too small. To check this, [15] investigated the alloy composition Ni_7Lu_2 and found that no phase with that composition exists, supporting the proposed theory. Thus, the stability of Ni_7Yb_2 remains in question, and it has not been included in Fig. 1 (dashed line). Thermal analysis by Palenzona and Cirafici [16] indicated a transformation in NiYb at 1000 °C. Buschow [15] as well as Palenzona and Cirafici [16] concluded from their works that no Ni–Yb intermediate phases are formed with compositions greater than 50 at.% Yb. This is explained on the basis that Yb is divalent at Yb-rich compositions and trivalent for Ni-rich

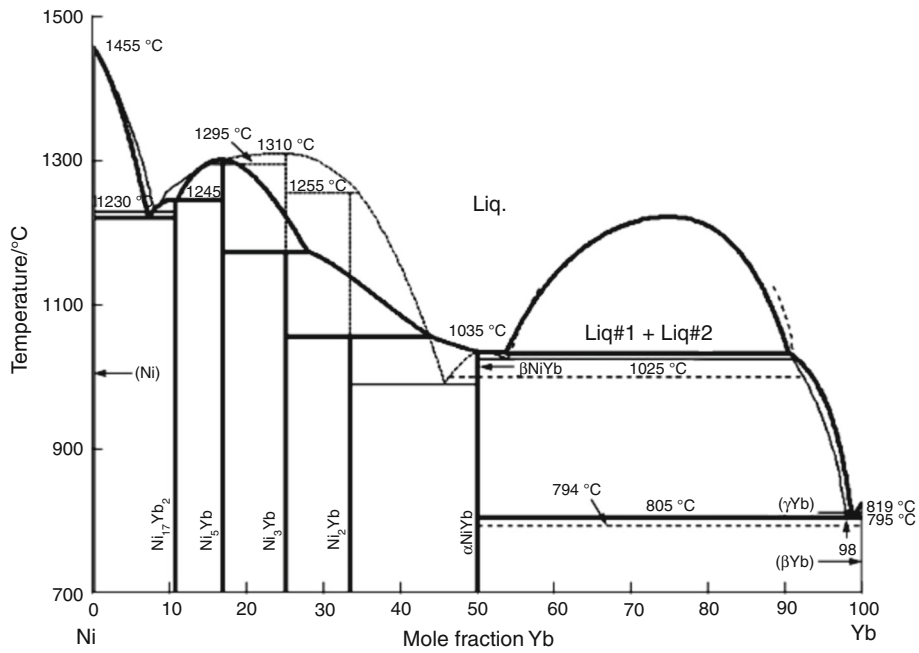

Fig. 1 Phase diagram of the Ni–Yb system, calculated by Berezutskii et al. [13] (*bold lines*) compared to the data from [16, 17, 19] (*dashed lines*)

compositions. Most recently, Berezutskii et al. [13] published an important study intended to determine whether the experimental mixing enthalpy of the liquid raises is consistent with the liquidus and the melting points of some compounds of both Ni–Yb and Ni–Eu systems. After exhibiting analysis of the phase diagrams of the Ni–Ln (Ln = La, Ce, Pr, Nd, Sm, Gd, Dy, Ho [19], Tb [20]) systems, Berezustkii et al. [13] suggested that the results of the study of Palenzona and Cirafici [16] are questionable. According to them, there are two more facts that stand against the congruent melting of Ni_3Yb at 1310 °C. Firstly, if this point takes such place on the phase diagram, the liquidus in the range $0.25 < x_{Ni} < 0.5$ must be very steep. Secondly, a problem of interaction of the Ni–Yb melts (at $x_{Yb} < 0.25$) with molybdenum crucibles was discussed by Nash [17]. It is known that lanthanides do not dissolve molybdenum, while nickel forms eutectic with $x_{Mo} = 0.33$, as well as the MoNi compound [Massalski 17]. Berezustkii et al. [13] reported that contamination of the alloys with molybdenum in the experiments by Palenzona and Cirafici [16] should be significant not only for $x_{Yb} < 0.25$, but also for $0.25 < x_{Yb} < 0.5$ alloys, because the nickel activity is still quite high in this range. So, the liquidus in the range $0.25 < x_{Yb} < 0.5$, determined by Palenzona et al. [16], may not be considered as reliable. Thus, the liquidus curve of the phase diagram of the Ni–Yb system is presented in Fig. 1 as a bold line. Hence, the most refractory compound in the Ni–Yb system should be Ni_5Yb.

Crystal structures

Crystallographic data of the Ni–Yb intermediate phases are given in Table 1. The crystal structures of NiYb and Ni_2Yb were confirmed by Ensslen et al. [21], although no details were given. The crystal structure of the high-temperature form of NiYb (NiYb_HT) observed by Palenzona and Cirafici [16] has not been determined.

Thermodynamic data

Bayanov et al. [22] calculated the enthalpy of formation of Ni_5Yb as -16.4 kJ mol^{-1} from a modified model of Kubaschewski [23]. This enthalpy value takes into account the energy involved in the transition from divalent Yb to trivalent Yb by Gschneidner et al. [24]. An explanation of the form of the phase diagram, with divalent Yb at Yb-rich compositions and trivalent Yb at Ni-rich compositions was given by Miedema [25]. This explanation shows that trivalent Yb has a more negative enthalpy than the divalent form, but when the transformation energy from divalent to trivalent is considered (33 kJ mol^{-1} [25]), the divalent form is more stable at Yb-rich compositions. Furthermore, the most stable configuration is the one in which pure, divalent Yb is in equilibrium with a Ni-trivalent Yb alloy, as it is observed experimentally by Nash [17]. Recently, Brutti et al. [26] investigated the thermochemistry of the Ni–Yb intermediate phases by means of tensiometric measurements and, on a few compositions, calorimetric

Table 1 Crystal structure data of stable solid phases in the Ni–Yb and Ni–Eu systems

Phase	Pearson symbol	Space group	Strukturbericht designation	Prototype	Phase modeling	Used models	References
Ni–Yb							
(Ni)	cF4	$Fm\bar{3}m$	*A1*	Cu	$(Ni,Yb)_1:(Va)_1$	SM (f)	[17, 18]
$Ni_{17}Yb_2$	(a)	$P6_3/mmc$	…	Th_2Ni_{17}	$(Ni)_{17}:(Yb)_2$	SCM (g)	[15, 16]
Ni_5Yb	hP6	$P6/mmm$	$D2_d$	$CaCu_5$	$(Ni)_5:(Yb)_1$	SCM	[15, 16, 59]
Ni_3Yb	(b)	$R\bar{3}m$	…	$PuNi_3$	$(Ni)_3:(Yb)_1$	SCM	[15, 16, 60]
Ni_2Yb	cF24	$Fd\bar{3}m$	*C15*	Cu_2Mg	$(Ni)_2:(Yb)_1$	SCM	[15, 16, 62]
α-NiYb	oP8	*Pnma*	*B27*	FeB	$(Ni)_1:(Yb)_1$	SCM	[15, 16]
β-NiYb	oP8	*Pnma*		FeB	$(Ni)_1:(Yb)_1$	SCM	
γYb (c)	cI2	$Im\bar{3}m$	*A2*	W	$(Ni,Yb)_1:(Va)_3$	SM	[18]
βYb (d)	cF4	$Fm\bar{3}m$	*A1*	Cu	$(Ni,Yb)_1:(Va)_1$	SM	[18]
αYb (e)	hP2	$P6_3/mmc$	*A3*	Mg	$(Ni,Yb)_1:(Va)_{0.5}$	SM	[18]
Ni–Eu							
(Ni)	cF4	$Fm\bar{3}m$	*A1*	Cu	$(Ni,Eu)_1:(Va)_1$	SM	[15]
$Ni_{17}Eu_2$	(a)	$P6_3/mmc$	…	Th_2Ni_{17}	$(Ni)_{17}:(Eu)_2$	SCM	[38–40]
Ni_5Eu	hP6	$P6/mmm$	$D2_d$	$CaCu_5$	$(Ni)_5:(Eu)$	SCM	[38, 40, 42, 43, 49, 61]
Ni_2Eu	cF24	$P6_3/mmc$	*C36*	Ni_2Mg	$(Ni)_2:(Eu)$	SCM	[47]
(Eu)	cI2	$Im\bar{3}m$	*A2*	W	$(Ni,Eu)_1:(Va)_3$	SM	[18]

(a) Hexagonal. (b) Rhombohedral. (c) From 795 to 819 °C. (d) From −3 to 795 °C. (e) <−3 °C. (f) SM, solution model. (g) SCM, stoichiometric compound model

measurements. In the latter study, the equilibrium vapor pressures over two solid phase regions were measured by Knudsen effusion mass spectrometry and Knudsen effusion mass loss, and the data were analyzed by the second and the third law methods in order to derive the enthalpy changes for the high-temperature decomposition reactions. Furthermore, the enthalpy of formation of the NiYb compound was preliminarily determined by direct reaction calorimetry (DRC). The heats of formation of the five intermediate phases were finally obtained by a combined analysis of all the collected data. Furthermore, Brutti et al. [26] measured vapor pressure equations $\log p_{Yb}/\text{bar} = -(A/T) + B$.

Moreover, a group of authors (Di Pascasio et al. [27], Cignini et al. [28] and Gozzi and Iervolino [29]) investigated the thermodynamic properties of the nickel-rich compounds with lanthanides, including $Ni_{17}Yb_2$ [27] and $Ni_{17}Eu_2$ [29] at $T = 990$ K and $T = 910$ K, respectively. The results obtained for the Ni–Yb system agree satisfactorily with those by Brutti et al. [26], as well as with some data for other Ni–Ln systems. In contrast, those obtained for the Ni–Eu system are dramatically different from what may be expected regarding the comparison with Ni–Ln systems, and this is noted by the authors [29] themselves, who did not find the explanation of such a fact. More recently, by using calorimetric technique, Berezutskii et al. [13] determined experimentally the mixing enthalpies of melts of the Ni–Yb and Ni–Eu systems and calculated the thermodynamic functions in wide ranges of temperature and concentration. They suggested a model where the excess mixing entropies of melts are proportional to the mixing enthalpies in the whole concentration range. Therefore, there is only one fitting parameter. According to them, the calculation of the phase diagrams of the Ni–Yb and Ni–Eu systems certainly needed the knowledge of the thermodynamic properties not only for liquid alloys but for solid phases too. They considered the Gibbs energies of formation of the solid phases as linearly dependent on temperature, i.e., $\Delta_f H(T) = \text{const}$, $\Delta_f S(T) = \text{const}$. Their values are listed and compared with those estimated by the Miedema's model [25] in Table 2.

The heat capacities of the compound (Ni_2Yb) in the temperature range from 4 to 300 K were determined by Rojas et al. [30] using a Quantum Design PPMS multipurpose instrument in the 2–300 K temperature range. The enthalpies of formation of all the binary intermetallic compounds in the Ni–Yb system available from the literature are given in Table 2.

Magnetic properties

The magnetic properties of several Ni–Yb intermediate phases were studied experimentally by Nesbitt et al. [31], Bleaney [32], Nowik et al. [33], Carfagna and Wallace

Table 2 Comparison of the measured, predicted and optimized enthalpies of formation of the intermetallic compounds of the Ni–Yb system

Intermetallic compounds	Enthalpy of formation/ kJ mol^{-1} at.$^{-1}$	Entropies of formation/ J mol^{-1} at.$^{-1}$ K^{-1}	Technique used	*T*/K	References
$Ni_{17}Yb_2$	−12.70	–	Prediction^{M}	298.15	[25]
	−13.9 ± 0.3	–	Vapor pressure	298.15	[26]
	−13.15	33.33	Optimization	298.15	Current work
	−14.80	−02.50	Calorimetry	298	[13]
	−17.0 ± 0.1	1.96 ± 0.07	Effusion and EMF^{a}	990	[27]
Ni_5Yb	−20.2	–	Prediction^{M}	298.15	[25]
	−16.4	–	Calculation	298.15	[22]
	−20.9 ± 4.4	–	Vapor pressure	298.15	[26]
	−20.47	33.50	Optimization	298.15	Current work
	−22.0	−04.00	Calorimetry	298	[13]
Ni_3Yb	−28.80	–	Prediction^{M}	298.15	[25]
	−26.0 ± 4.8	–	Vapor pressure	298.15	[26]
	−23.42	33.98	Optimization	298.15	Current work
	−24.80	−06.00	Calorimetry	298	[13]
Ni_2Yb	−33.60	–	Prediction^{M}	298.15	[25]
	−30	–	Calorimetry	298.15	[26]
	−32.0 ± 4.6	–	Vapor pressure	298.15	[26]
	−22.07	36.47	Optimization	298.15	Current work
	−23.30	−05.30	Calorimetry	298	[13]
NiYb_LT	−27.80	–	Prediction^{M}	298.15	[25]
	−28.0 ± 2.0	–	Synthèse directe Calo.	298.15	[26]
	−17.28	42.43	Optimization	298.15	Current work
	−18.80	−04.00	Calorimetry	298	[13]

$^{R–K}$ Issue from Redlich–Kister equation, ^{M} Issue from Miedema model

[34], Nowik and Dunlap [35], Klaasse et al. [36], and theoretically by Bleaney [37].

The Ni–Eu system

Phase diagram data

Savitskii et al. [38] using thermal analysis, metallography, hardness measurements and X-ray diffractography have investigated the phase equilibria. Ten alloys containing 0–4 at.% Eu and one alloy of about 19 at.% Eu were prepared from electrolytic Ni and Eu containing <0.23 mass % impurities. Two compounds were established in this composition range $Ni_{17}Eu_2$ and Ni_5Eu. The proposed peritectic formation of $Ni_{17}Eu_2$ was not documented by thermal or metallographic evidence, nor even discussed by Savitskii et al. [38]. On the basis that the alloy containing about 19 at.% Eu melted at 1440 ± 10 °C, [38] assumed that Ni_5Eu melts congruently near this temperature. Prior to the work of [38], Chechernikov et al. [39] used electrolytic Ni (99.9 at.% pure) and Eu containing less than 0.1 at.% Sm and Gd, 0.075 at.% Ca, 0.01 at.% Fe, 0.005 at.% Cu and 0.04 at.% Ta and indicated a eutectic reaction Liq ↔ (Ni) + $Ni_{17}Eu_2$ at 1190 ± 10 °C compared with 1200 ± 10 °C [38]. The eutectic composition was estimated to be about 5 to 6 at.% Eu. The liquidus between 0 and 4 at.% Eu in this system is established and is shown with a solid line; the remainder is not determined. Alloys with 0.3 at.% Eu or more quenched from 1100 °C had a (Ni) lattice parameter of 0.35216 nm, compared with the value 0.35211 nm for pure Ni [38], i.e., an expansion of 0.014 %. The maximum solubility of Eu in (Ni) therefore is concluded to be less than 0.3 at.% Eu at 1100 °C decreasing to essentially zero at room temperature. Chechernikov et al. [39] performed X-ray diffraction and microstructure analyses and obtained the same results. They determined the

highest Ni compound in this system to be $Ni_{17}Eu_2$, which was confirmed by Savitskii et al. [38] and Terekhova et al. [40]. Ni_5Eu was reported first by Kripyakevich and Gladishevs'kii [41] and confirmed by Palenzona [42], Savitskii et al. [38] and Terekhova [40]. A compilation of lattice parameters of Ni_5Eu in Gavra et al. [43] showed differences among investigators that might suggest that Ni_5Eu exists over a range of solid solubility. Efforts to produce Ni2Eu by Moriarty et al. [44] were unsuccessful. Olcese [45] in his study of the RET systems (where T is an element of the 8th group) did not find Ni_3Eu_7. The results were taken by Tung and Nash [46] to draw an assessed partial phase diagram. Ning et al. [47] have found an additional intermediate phase, Ni2Eu. Thus, Okamoto [48] has drawn speculatively the phase diagram for the whole concentration range. This tentative phase diagram was taken to construct Fig. 2 (dashed line). The peritectic formation of Ni_2Eu is based on assumed similarity to the Ni–Yb system. Ni_3Yb exists in the Ni–Yb system, but the existence of Ni_3Eu has not been reported. Probably, Ni_3Eu is metastable. The region between Ni2Eu and (Eu) is tentatively drawn assuming similarity to the Ni–Yb system. The existence of NiEu and the liquid immiscibility needs confirmation. The initial slope of (Ni) liquidus is modified in Fig. 2 (dashed line) because, according to [49], it was too steep in the diagram of Tung et al. [46]. In the phase diagram published by Berezutskii et al. [13], there are two points that we remarked:

- It appears that four phases occur in the invariant equilibrium at 1275 °C: fcc_A1(Ni), Liq, Eu_2Ni_{17} and $EuNi_5$ which is in violation with the Gibb's phase rule ($F = C + 1 - P$, with $F = 0$ for an invariant equilibrium, $C =$ two constituents Ni and Eu, 1 for the temperature as the pressure is constant = 1e5 Pa, therefore the number of phases $P = 3$). We think there is the eutectic reaction Liquid ↔ fcc_A1 (Ni) + Eu_2Ni_{17} and a peritectic reaction Liq + $EuNi_5$ ↔ Eu_2Ni_{17}. The temperatures of the two invariant reactions may differ by about 1 °C.
- In the reaction at 1095 °C, the monotectic reaction: Liq (about 50.1 at.% Eu) ↔ EuNi (50 at.% Eu) + Liquid#2 (about 95 at.% Eu) may have a temperature, that is, about 1 or 2 °C lower that the congruent melting of the EuNi phase.

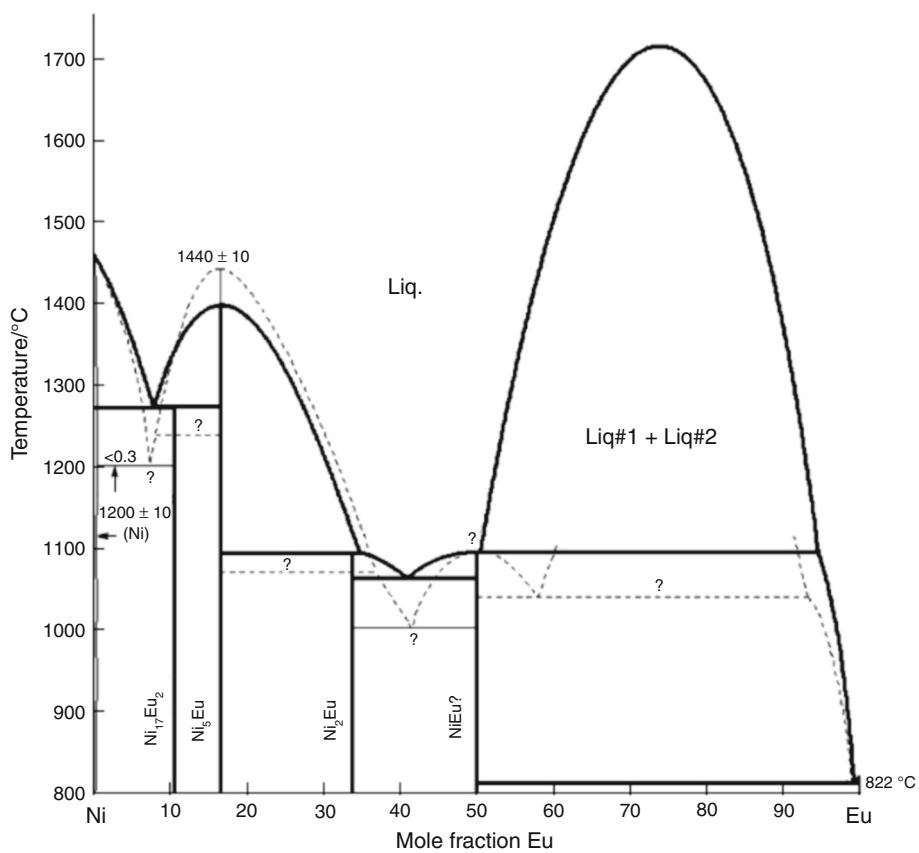

Fig. 2 Phase diagram of the Ni–Eu system, calculated by Berezutskii et al. [13] (*bold lines*) compared to the data [48] (the symbol ? means uncertain, estimated temperatures of the phase transformations as *dashed lines*)

Table 3 Optimized thermodynamic parameters of the Ni–Yb system (J mol^{-1} at^{-1} and J mol^{-1} at^{-1} K^{-1})

Phases	Thermodynamic models	Optimization, Redlich–Kister equation [14]
Liquid	(Ni,Yb)	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ni:Yb}} = -28520 + 1.85 * T$
		${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ni:Yb}} = -56380 + 3.66 * T$
		${}^{2}L^{\mathrm{Liq}}_{\mathrm{Ni:Yb}} = -34790 + 2.26 * T$
		${}^{3}L^{\mathrm{Liq}}_{\mathrm{Ni:Yb}} = -13028.9 + 0.96 * T$
		${}^{4}L^{\mathrm{Liq}}_{\mathrm{Ni:Yb}} = -7.85 - 0.52 * T$
FCC_A1	$(\mathrm{Ni,Yb})_1:(\mathrm{Va})^{a}_{1}$	${}^{0}L^{\mathrm{FCC_A1}}_{\mathrm{Ni,Yb:Va}} = 2E + 05$
BCC_A2	$(\mathrm{Ni,Yb})_1:(\mathrm{Va})^{a}_{3}$	${}^{0}L^{\mathrm{BCC_A2}}_{\mathrm{Ni,Yb:Va}} = 5E + 04$
HCP_A3	$(\mathrm{Ni,Yb})_1:(\mathrm{Va})^{a}_{0.5}$	No excess term
$Ni_{17}Yb_2$	$Ni_{0.895}:Yb_{0.105}$	$G^{\mathrm{Ni_{17}Yb_2}}_{\mathrm{Ni:Yb}} - 0.895 * H^{298.15}_{\mathrm{Ni}} - 0.105 * H^{298.15}_{\mathrm{Yb}} = 0.895 * \mathrm{GHSERNi} + 0.105 * \mathrm{GHSERYb} - 14800 + 2.5 * T$
Ni_5Yb	$Ni_{0.833}:Yb_{0.167}$	$G^{\mathrm{Ni_5Yb}}_{\mathrm{Ni:Yb}} - 0.833 * H^{298.15}_{\mathrm{Ni}} - 0.167 * H^{298.15}_{\mathrm{Yb}} = 0.833 * \mathrm{GHSERNi} + 0.167 * \mathrm{GHSERYb} - 22200 + 4.0 * T$
Ni_3Yb	$Ni_{0.750}:Yb_{0.250}$	$G^{\mathrm{Ni_3Yb}}_{\mathrm{Ni:Yb}} - 0.750 * H^{298.15}_{\mathrm{Ni}} - 0.250 * H^{298.15}_{\mathrm{Yb}} = 0.750 * \mathrm{GHSERNi} + 0.250 * \mathrm{GHSERYb} - 24800 + 5.74 * T$
Ni_2Yb	$Ni_{0.667}:Yb_{0.333}$	$G^{\mathrm{Ni_2Yb}}_{\mathrm{Ni:Yb}} - 0.667 * H^{298.15}_{\mathrm{Ni}} - 0.333 * H^{298.15}_{\mathrm{Yb}} = 0.667 * \mathrm{GHSERNi} + 0.333 * \mathrm{GHSERYb} - 23300 + 5.48 * T$
NiYb_LT	$Ni_{0.500}:Yb_{0.500}$	$G^{\mathrm{NiYb_LT}}_{\mathrm{Ni:Yb}} - 0.500 * H^{298.15}_{\mathrm{Ni}} - 0.500 * H^{298.15}_{\mathrm{Yb}} = 0.500 * \mathrm{GHSERNi} + 0.500 * \mathrm{GHSERYb} - 18200 + 4.0 * T$

$(\mathrm{Va})^{a}$ for vacancy. The exponent ^{a} depends on the crystalline structure of the phase (bcc_A2, fcc_A1 and hcp_A3)

Table 4 Comparison of the optimized enthalpies and entropies f formation of the intermetallic compounds of the Ni–Yb, Ni–Eu, Ni–Ho and Ni–Sc systems

Intermetallic compounds	Enthalpy of formation/ kJ mol^{-1} at.$^{-1}$	Entropies of formation/ J mol^{-1} at.$^{-1}$ K^{-1}	Technique used	*T*/K	References
Ni17Yb2	−13.15	33.33	Optimization	298.15	Present work
Ni17Ho2	−21.33	33.56	Optimization	298.15	[12]
Ni17Eu2	−9.00	37.78	Optimization	298.15	Present work
Ni5Yb	−20.47	33.50	Optimization	298.15	Present work
Ni5Ho	−34.04	34.27	Optimization	298.15	[12]
Ni5Eu	−14.47	41.19	Optimization	298.15	Present work
Ni5Sc	−17.59	32.52	Optimization	298.15	[11]
Ni2Yb	−22.70	36.47	Optimization	298.15	Present work
Ni2Ho	−42.67	42.32	Optimization	298.15	[12]
Ni2Eu	−16.36	49.55	Optimization	298.15	Present work
Ni2Sc	−33.77	29.38	Optimization	298.15	[11]
NiYb	−17.28	42.43	Optimization	298.15	Present work
NiHo	−48.23	46.21	Optimization	298.15	[12]
NiEu	−16.22	57.74	Optimization	298.15	Present work
NiSc	−45.67	23.12	Optimization	298.15	[11]

Crystal structures

Ni–Eu crystal structure data are listed in Table 1. Pure Eu metal has an atomic volume of 28.979 cm^3/mol, a density calculated from X-ray data of about 5.244 g/cm^3 and a mean metallic radius of 0.20418 nm [18]. Oliver et al. [49] found Eu in Ni_5Eu to be trivalent. Due to the chemical similarity between Ca and Eu, the intermetallic compounds $CaNi_5$ and Ni_5Eu have the same structure and cell volume and exhibit similar chemistry and physical properties [43].

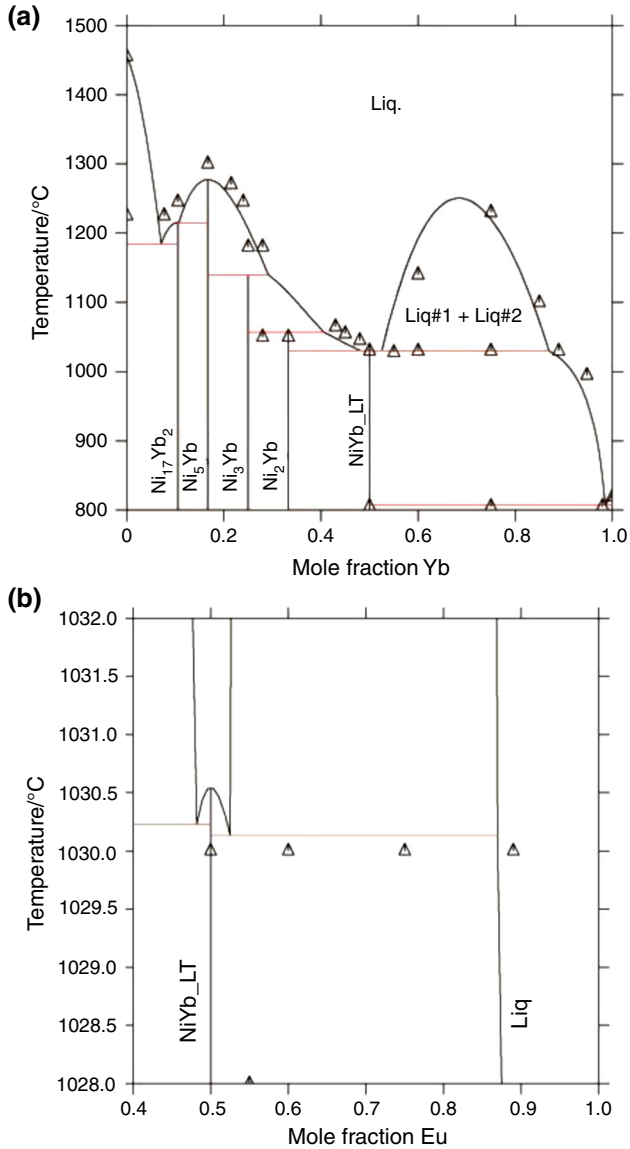

Fig. 3 **a** Calculated phase diagram Ni–Yb during this work compared to the experimental data (*triangle*) by Nash [17]. **b** Zoom of the calculated Ni–Yb phase diagram around $x(\mathrm{Yb}) = 0.5$ at

Magnetic properties

According to Tung and Nash [46], Chechernikov et al. [39] studied the magnetic properties of Ni–Eu alloys between 300 and 1000 °C. The reciprocal value of the specific susceptibility $1/\chi$ is linear in a wide temperature range. Only at very high temperatures does it start to level off, and the same is found for pure Ni. $Ni_{17}Eu_2$ was also produced, but because it contained up to 1.0 at.% Mo, and its magnetic properties were not studied by Chechernikov et al. [39]. The compound Ni_5Eu did not show any evidence of magnetic ordering down to 2.6 K [49].

Thermodynamic models

Pure elements

The Gibbs energy function $G_i^{\Phi}(T) = {}^0G_i^{\mathrm{SER}}(T) - {}^0H_i^{\mathrm{SER}}(298.15\ \mathrm{K})$ for the element i (i = Ni, Yb or Eu) in the phase Φ (Φ = liquid, body-centered cubic (bcc_A2), face-centered cubic (fcc_A1) or hexagonal close-packed (hcp_A3)) is described by an equation of the following form:

$$G_i^{\Phi}(T) = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

where ${}^0H_i^{\mathrm{SER}}(298.15\ \mathrm{K})$ is the molar enthalpy of the element i at 298.15 K and 1 bar in its standard element reference (SER) state, fcc_A1 for Ni, hcp_A3 for Yb and bcc_A2 for Eu.

In the present work, the Gibbs energy functions are taken from the SGTE (Scientific Group Thermodata Europe) compilation by Dinsdale [50].

Solution phases

In the Ni–Yb and Ni–Eu binary systems, the Gibbs free energies of the liquid, hcp_A3, bcc_A2 and fcc_A1 phases are described by the subregular solution model with the Redlich–Kister formula [14]. The Gibbs energy of one mole of formula unit is expressed as the sum of four terms:

$$G^{\Phi} - H_{\mathrm{SER}}^{\Phi} = {}^{\mathrm{ref}}G^{\Phi} + {}^{\mathrm{id}}G^{\Phi} + {}^{\mathrm{ex}}G^{\Phi} + {}^{\mathrm{mag}}G^{\Phi} \quad (2)$$

$${}^{\mathrm{ref}}G^{\Phi} = x\left[{}^0G_{\mathrm{Ni}}^{\Phi} - H_{\mathrm{Ni}}^{\mathrm{SER}}\right] + x_{\mathrm{RE}}\left[{}^0G_{\mathrm{RE}}^{\Phi} - H_{\mathrm{RE}}^{\mathrm{SER}}\right] \quad (3)$$

$${}^{\mathrm{id}}G^{\Phi} = RT(x_{\mathrm{Ni}}\ln x_{\mathrm{Ni}} + x_{\mathrm{RE}}\ln x_{\mathrm{RE}}) \quad (4)$$

$${}^{\mathrm{mag}}G^{\Phi} = RT\ln(\beta_O + 1)g(\tau) \quad (5)$$

where RE = Yb or Eu; R is the ideal gas constant; T is the absolute temperature, in Kelvin; x_{Ni} and x_{RE} are the mole fraction of the elements Ni and RE, respectively; $\tau = T/T^*$ with T^* is the critical temperature (the Curie temperature T_{C} for the ferromagnetic phase bcc_A2 and the Néel temperature T_{N} for the fcc_A1 phase); β_0 is the average magnetic moment per atom and $g(\tau)$ depends of τ [52].

Table 5 Invariant reactions in the Ni–Yb system

Reactions	*T*/°C	Reaction type	*x*(Liq,Yb) *x*(Liq1,Yb) *x*(Liq2,Yb)	Methods	References
Liq ↔ fcc_A1 + Ni17Yb2	1230	Eutectic	0.080	XRD/DTA/metallography	[16, 17]
	1222	Eutectic	0.073	Calculated	[13]
	1185	Eutectic	0.072	Calculated	This work
Liq + Ni5Yb ↔ Ni17Yb2	1245	Peritectic	0.085	XRD/DTA/metallography	[16, 17]
	1245	Peritectic	0.105	Calculated	[13]
	1215	Peritectic	0.107	Calculated	This work
Liq ↔ Ni5Yb	–	–	–	XRD/DTA/metallography	[16, 17]
	1300	Congruent	0.166	Calculated	[13]
	1277	Congruent	0.167	Calculated	This work
Liq + Ni3Yb ↔ Ni5Yb	1295	Peritectic	0.152	XRD/DTA/metallography	[16, 17]
	1173	Peritectic	0.280	Calculated	[13]
	1140?	Peritectic	0.205	Calculated	This work
Liq ↔ Ni3Yb	1310	–	0.250	XRD/DTA/metallography	[16, 17]
	–	Congruent	–	Calculated	[13]
	–	Congruent	–	Calculated	This work
Liq + Ni3Yb ↔ Ni2Yb	1255	–	–	XRD/DTA/metallography	[16, 17]
	1057	Peritectic	0.436	Calculated	[13]
	1057	Peritectic	0.406	Calculated	This work
Liq ↔ NiYb_HT	1035	–	0.50	XRD/DTA/metallography	[16, 17]
	1035	Congruent	0.50	Calculated	[13]
	–	Congruent	0.50	Calculated	This work
Ni2Yb + NiYb_LT ↔ Liq	990	Eutectic	0.455	XRD/DTA/metallography	[16, 17]
	–	–	–	Calculated	[13]
	1030	Eutectic	0.482	Calculated	This work
NiYb_LT ↔ NiYb_HT	1000	Polymorphic	–	XRD/DTA/metallography	[16, 17]
	–	–	–	Calculated	[13]
	1030.5	Polymorphic	0.500	Calculated	This work
Liq1#↔ Liq2# + NiYb_LT	1025	Eutectic	0.536–0.909	XRD/DTA/metallography	[16, 17]
	1033	Eutectic	0.536–0.909	Calculated	[13]
	1030	Eutectic	0.527–0.870	Calculated	This work
Top of the monotectic domain	1230	Monotectic	0.75	Calculated	[13]
Top of the monotectic domain	1249	Monotectic	0.675	Calculated	This work
Liq↔NiYb_LT + fcc_A1	794	Eutectic	0.980	XRD/DTA/metallography	[16, 17]
	805	Eutectic	0.981	Calculated	[13]
	807	Eutectic	0.986	Calculated	This work

Comparison between the experimental data and the calculated values after the optimizations using the Redlich–Kister formalism [14]

$$^{ex}G^{\Phi} = x_i x_j \sum_{\lambda=0}^{n} {}^{\lambda}L^{\Phi}_{i,j}(x_i - x_j)^{\lambda} \quad (6)$$

x_i and x_j denote the site fraction in the sublattice, *i* and *j* the indices which correspond to the species Ni and RE and $x_i + x_j = 1$.

$^{\lambda}L^{\Phi}_{i,j}$ ($\lambda = 0, 1, 2,\ldots$) is the *i*th interaction parameter between the elements Ni and RE and to be evaluated in the present work, expressed as:

$$^{\lambda}L^{\Phi}_{i,j} = a_{\lambda} + b_{\lambda}T \quad (7)$$

additional terms can be added if needed to describe the temperature dependence of the experimental data. An important step in the optimization procedure is the selection of which and how many of the adjustable coefficients shall be used for the evaluation. By using too many coefficients, the resulting functions may follow the scatter of the experimental data and thus lead to unexpected curvatures. An insufficient number of coefficients, however, may result in an inaccurate description of the thermodynamic functions.

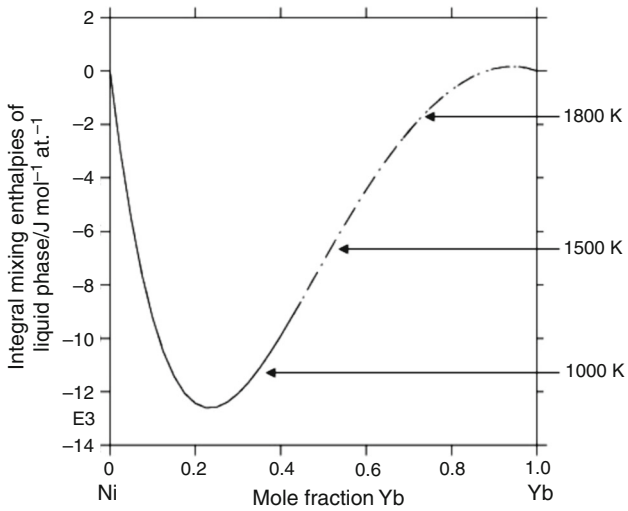

Fig. 4 Integral enthalpies of mixing of the Ni–Yb liquid phase calculated at 1000 K (*continuous line*), 1500 K (*dashed line*) and 1800 K (*dotted line*)

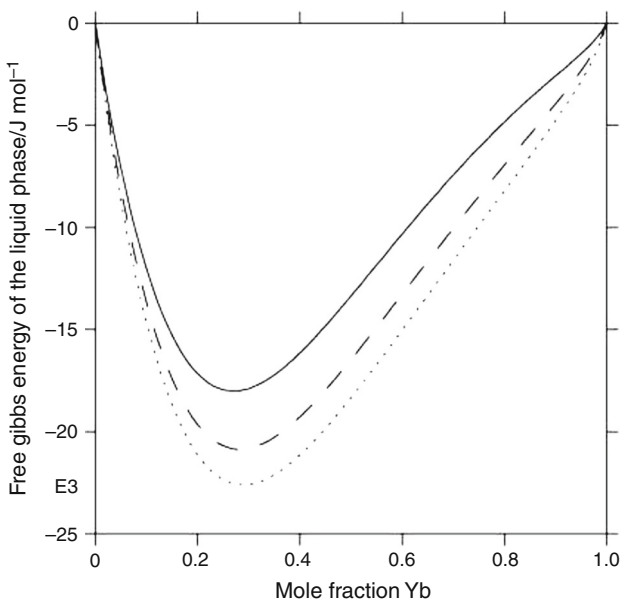

Fig. 5 Free Gibbs energy of mixing of the Ni–Yb liquid phase calculated at 1000 K (*continuous line*), 1500 K (*dashed line*) and 1800 K (*dotted line*)

The magnetic formalism was defined by Inden [51, 52] and Hillert and Jarl [53] and included in Dinsdale paper [50]. In this paper, the value of the p parameter in $g(\tau)$, which can be thought as the fraction of the magnetic enthalpy absorbed above the critical temperature (T_C), depends of the crystal structure of the phase. For the bcc_A2 phase $p = 0.4$, and for the other common phases $p = 0.28$.

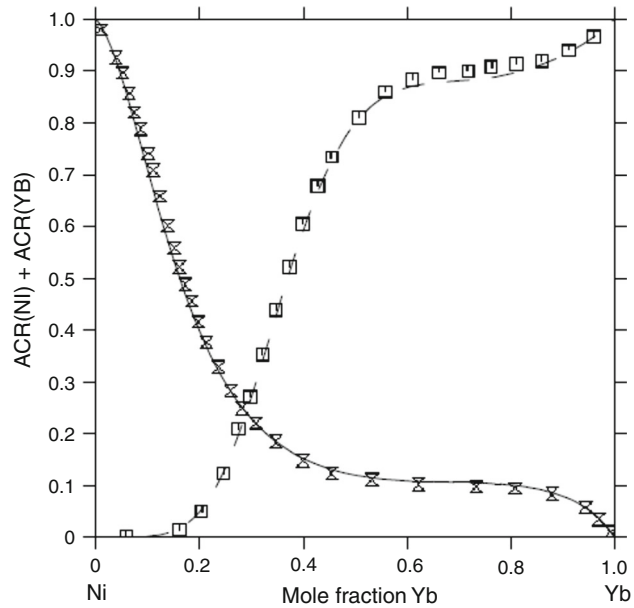

Fig. 6 Calculated thermodynamic activities for the liquid phase [Ni in Yb (*solid line*) and for Yb in Ni (*dashed line*)] at $T = 1600$ K in Ni–Yb system compared with experimental ones by Berezutskii et al. [13] (⋈, *square*)

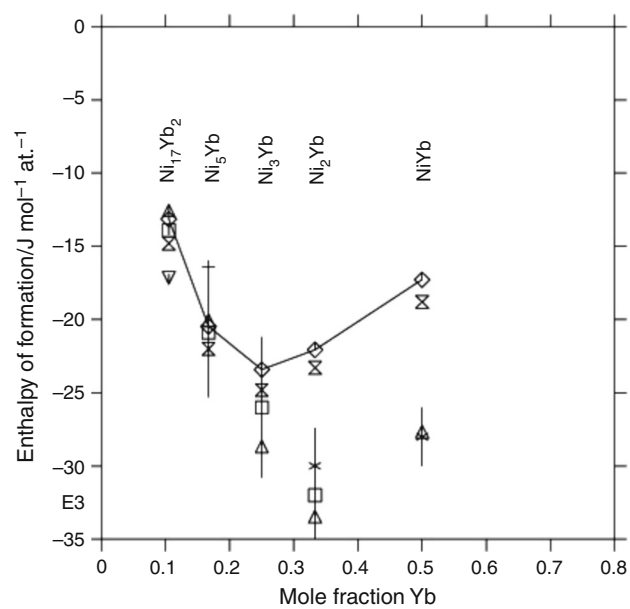

Fig. 7 Calculated and predicted enthalpies of formation of the Ni–Yb compounds compared with various experimental data, (| is the error bar), *diamond* this work, *triangle* Miedema [25], *inverted triangle* Pascasio [27], *square*, x Brutti et al. [26], Berezutskii et al. [13], *plus symbol* Bayanov [22]

Stoichiometric compounds

The intermetallic compounds in the Ni–Eu and Ni–Yb systems, including $Ni_{l7}Yb_2$, Ni_5Yb, Ni_3Yb, Ni_2Yb, αNiYb,

Table 6 Optimized thermodynamic parameters of the Ni–Eu system (J mol^{-1} at.$^{-1}$ and J mol^{-1} at.$^{-1}$ K^{-1})

Phases	Thermodynamic models	Optimization, Redlich–Kister equation [14]
Liquid	(Ni,Eu)	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ni:Eu}} = -22750 - 14.58 * T$
		${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ni:Eu}} = -39993 + 7.915 * T$
		${}^{2}L^{\mathrm{Liq}}_{\mathrm{Ni:Eu}} = -17439 + 12.22 * T$
		${}^{3}L^{\mathrm{Liq}}_{\mathrm{Ni:Eu}} = -28541 - 3.44 * T$
FCC_A1	$(\mathrm{Ni,Eu})_1:(\mathrm{Va})^{a}_{1}$	${}^{0}L^{\mathrm{FCC_A1}}_{\mathrm{Ni,Eu:Va}} = 2E + 05$
		${}^{0}T_{\mathrm{C}}{}^{\mathrm{FCC_A1}}_{\mathrm{Ni:Va}} = 633$ K
		${}^{0}\beta_{\mathrm{mag}}{}^{\mathrm{FCC_A1}}_{\mathrm{Ni:Va}} = 0.52$
BCC_A2	$(\mathrm{Ni,Eu})_1:(\mathrm{Va})^{a}_{3}$	${}^{0}L^{\mathrm{BCC_A2}}_{\mathrm{Ni,Eu:Va}} = 5E + 04$
		${}^{0}T_{\mathrm{C}}{}^{\mathrm{BCC_A1}}_{\mathrm{Ni:Va}} = 575$ K
		${}^{0}\beta_{\mathrm{mag}}{}^{\mathrm{BCC_A1}}_{\mathrm{Ni:Va}} = 0.85$
$Ni_{17}Eu_2$	$Ni_{0.895}:Eu_{0.105}$	$G^{\mathrm{Ni_{17}Eu_2}}_{\mathrm{Ni:Eu}} - 0.895 * H^{298.15}_{\mathrm{Ni}} - 0.105 * H^{298.15}_{\mathrm{Eu}} = 0.895 * \mathrm{GHSERNi} + 0.105 * \mathrm{GHSEREu} - 10650 + 0.25 * T$
Ni_5Eu	$Ni_{0.833}:Eu_{0.167}$	$G^{\mathrm{Ni_5Eu}}_{\mathrm{Ni:Eu}} - 0.833 * H^{298.15}_{\mathrm{Ni}} - 0.167 * H^{298.15}_{\mathrm{Eu}} = 0.833 * \mathrm{GHSERNi} + 0.167 * \mathrm{GHSEREu} - 16000 - 0.192 * T$
Ni_2Eu	$Ni_{0.667}:Eu_{0.333}$	$G^{\mathrm{Ni_2Eu}}_{\mathrm{Ni:Eu}} - 0.667 * H^{298.15}_{\mathrm{Ni}} - 0.333 * H^{298.15}_{\mathrm{Eu}} = 0.667 * \mathrm{GHSERNi} + 0.333 * \mathrm{GHSEREu} - 17586.3 - 0.618 * T$
NiEu	$Ni_{0.500}:Eu_{0.500}$	$G^{\mathrm{NiEu}}_{\mathrm{Ni:Eu}} - 0.500 * H^{298.15}_{\mathrm{Ni}} - 0.500 * H^{298.15}_{\mathrm{Eu}} = 0.500 * \mathrm{GHSERNi} + 0.500 * \mathrm{GHSEREu} - 17139 - 0.8325 * T$

$(\mathrm{Va})^{a}$ for vacancy. The exponent ^{a} depends on the crystalline structure of the phase (bcc_A2, fcc_A1 and hcp_A3)

βNiYb, $Ni_{17}Eu_2$, Ni_5Eu, Ni_2Eu and NiEu, are treated as stoichiometric compounds. The Gibbs energy of the Ni_pRE_q stoichiometric compounds denoted as ${}^{0}G_{\mathrm{Ni_pRE_q}}$ is expressed as follows:

$$ {}^{0}G_{\mathrm{A_pB_q}} = \frac{p}{p+q}\,{}^{0}G_{\mathrm{A}} + \frac{q}{p+q}\,{}^{0}G_{\mathrm{B}} + \Delta G^{\mathrm{f}}_{\mathrm{A_pB_q}} + {}^{\mathrm{mag}}G_{\mathrm{A_pB_q}} \tag{8} $$

where ${}^{0}G_{\mathrm{A}}$ and ${}^{0}G_{\mathrm{B}}$ are the Gibbs energy of the pure elements Ni and RE, respectively; $\Delta G^{\mathrm{f}}_{\mathrm{A_pB_q}}$ is the Gibbs energy of formation per mole of formula unit Ni_pRE_q.

Because of a lack of experimental measurements, it is assumed that the Neumann–Kopp [54] rule applies to the heat capacity, i.e.,$\Delta C_{\mathrm{p}} = 0$. Thus, $\Delta G^{\mathrm{f}}_{\mathrm{Ni_pRE_q}}$ can be given by the following expression:

$$ \Delta G^{\mathrm{f}}_{\mathrm{A_pB_q}} = a + b \times T \tag{9} $$

where the parameters a and b were evaluated in the present work. ${}^{\mathrm{mag}}G_{\mathrm{A_pB_q}}$ is the magnetic contribution to the Gibbs energy, which is discussed in "Solution phases" section.

Because the Curie temperatures of all the compounds in the Ni–Yb and Ni–Eu systems are much lower than 298.15 K, the compounds are in a paramagnetic state above 298.15 K. Furthermore, the thermodynamic functions of the Ni, Yb and Eu elements are given in a temperature range above 298.15 K [50]. So, we did not take into account the magnetic contribution to the Gibbs energy of the intermetallic compounds during the optimization procedure.

Optimization procedure

The thermodynamic optimization of the model parameters of the Gibbs energy expressions is an application of the CALPHAD technique with the help of the PARROT module of the Thermo-Calc software developed by Jansson [55] and Sundman et al. [56]. The program works by minimizing the squared sum of the differences between measured and calculated values. During the assessment, most of the experimental data mentioned in "Review on literature data" section were given a certain mass. The mass is chosen by personal judgment and changed by trial and error during the work until most of the selected experimental information is reproduced within the expected uncertainty limits. In order to avoid the appearance of an unwanted inverted miscibility gap in the liquid phase, thermodynamic constraints were imposed during the optimization with the Redlich–Kister formalism [14]; i.e., a positive curvature of the liquidus [57, 58] by imposing the

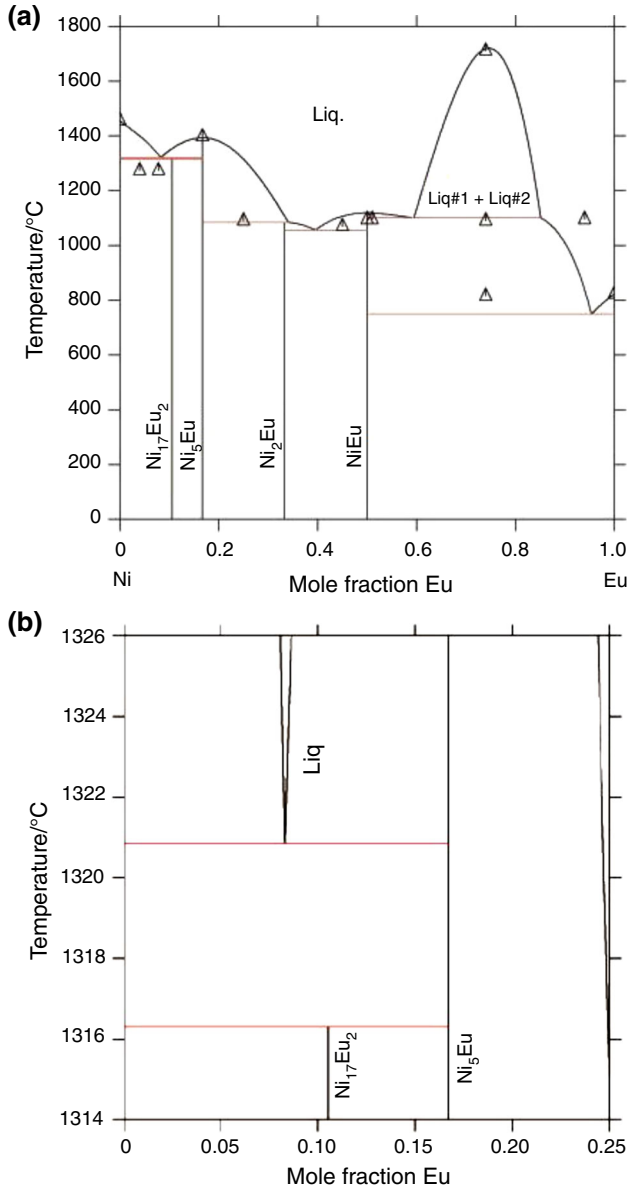

Fig. 8 **a** Calculated Ni–Eu phase diagram during this work compared with the experimental data (*triangle*) by Berezutskii et al. [13]. **b** Zoom of the calculated Ni–Eu phase diagram in the Ni-rich corner

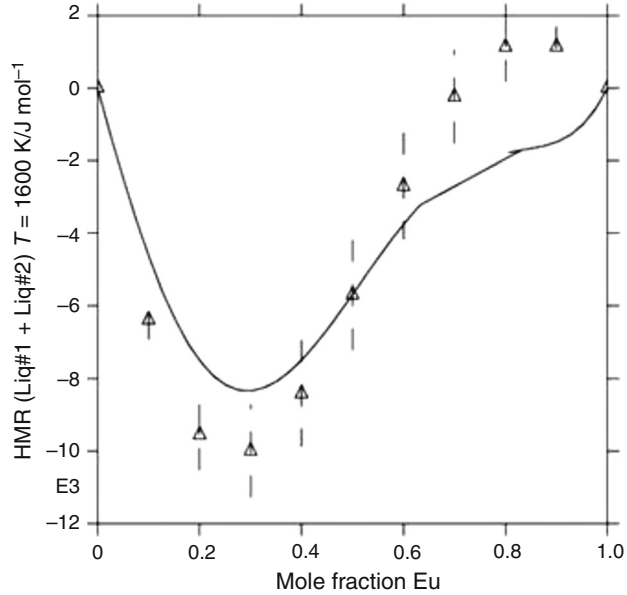

Fig. 9 Calculated enthalpies of mixing of the liquid phase at 1600 K in Ni–Eu binary system compared with those measured by Berezutskii et al. [13]

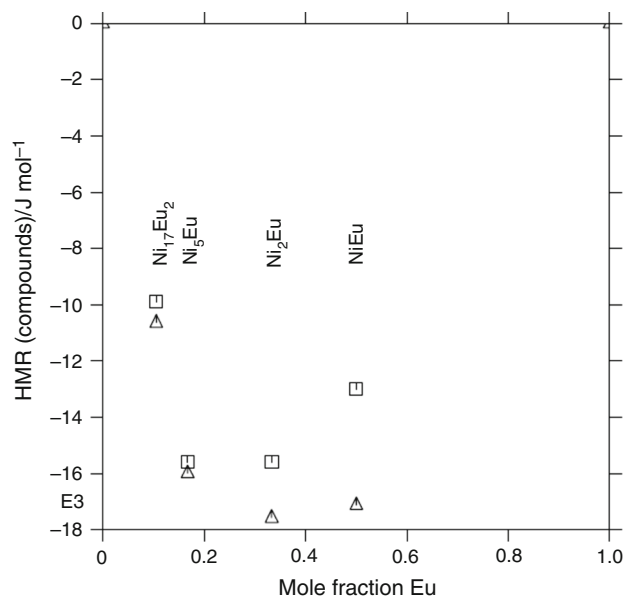

Fig. 10 Calculated enthalpies of formation (*triangle*) for the intermetallic compounds in the Ni–Eu phase diagram compared to the experimental ones (*square*) of [13]

additional constraint $d^2G/dx^2 > 0$ in the atomic composition range $0 \leq x_{RE(RE=Yb,Eu)} \leq 1$ and every 50° from the liquidus temperature, up to 6000 K, was therefore optimized. We precised that with our Thermo-Calc software, the excess parameters of the liquid phase are optimized with the Parrot module differently compared with the work of Berezustkii et al. [13]. The excess enthalpy and entropy are separately optimized. No linear dependence between them is used as in the work of [13].

Calculated results and discussion

Ni–Yb system

In Table 3, the optimized thermodynamic parameters of the Ni–Yb system during this work are listed. In Table 2 where the predicted, measured enthalpies and entropies of

Table 7 Comparison of the measured and optimized enthalpies of formation of the intermetallic compounds of the Ni–Eu system

Intermetallic compounds	Enthalpy of formation/ kJ mol^{-1} at.$^{-1}$	Entropies of formation/ J mol^{-1} at.$^{-1}$ K^{-1}	Technique used	*T*/K	References
Ni17Eu2	−9.0	37.79	Optimization[R–K]	298.15	Current work
	−9.9	−0.1	Calorimetry	298	[13]
	−48.0 ± 4	2.0 ± 0.2	Effusion and EMF[a]	910	[28]
Ni5Eu	−14.47	41.19	Optimization[R–K]	298.15	Current work
	−15.6	−0.10	Calorimetry	298	[13]
Ni2Eu	−16.36	49.55	Optimization[R–K]	298.15	Current work
	−15.6	−0.2	Calorimetry	298	[13]
	−16.22	57.74	Optimization[R–K]	298.15	Current work
NiEu	−13.0	−02.00	Calorimetry	298	[13]

[R–K] Issue from Redlich–Kister equation, [a]EMF electro motive force

Table 8 Invariant reactions in the Ni–Eu system

Invariant reaction	*T*/ °C	Reaction type	*x*(Liq,Eu) *x*(Liq1, Eu) *x*(Liq2, Eu)	Methods	References
Liq↔ fcc_A1 + Ni5Eu	1200 ± 10	Eutectic	0.05	XRD/DTA/metallography	[38]
	1200 ± 10		0.073	Assessed	[46]
	1270		0.078	Calculated	[13]
	1320.8		0.0827	Calculated	This work
Liq + Ni5Eu ↔ Ni17Eu2	1270 ± 1	Peritectic	0.078	XRD/DTA/metallography	[38, 46]
	1237		0.0788	Calculated	[13]
	1316.3		0.0827	Calculated	This work
Liq ↔ Ni5Eu	1440 ± 10	Congruent		XRD/DTA/metallography	[38, 46]
	1397		0.166	Calculated	[13]
	1393			Calculated	This work
Liq + Ni5Eu ↔ Ni2Eu	1069.5	Peritectic	0.366	XRD/DTA/metallography	[38, 46]
	1092.4		0.347	Calculated	[13]
	1085		0.341	Calculated	This work
Liq ↔ NiEu	1092.4	Congruent	0.5	XRD/DTA/metallography	[38, 46]
	1092.4		0.5	Calculated	[13]
	1118.6		0.5	Calculated	This work
Ni2Eu + NiEu ↔ Liq	1008	Eutectic	0.415	XRD/DTA/metallography	[38, 46]
	1061.9		0.410	Calculated	[13]
	1057		0.396	Calculated	This work
Liq1# ↔ Liq2# + NiEu	1039	Eutectic	0.581–0.934	XRD/DTA/metallography	[38, 46]
	1092.4	Eutectic	0.505–0.948	Calculated	[13]
	1098	Eutectic	0.600–0.851	Calculated	This work
Top of the monotectic domain	1710	Monotectic	0.74	Calculated	[13]
Top of the monotectic domain	1720	Monotectic	0.745	Calculated	This work
Liq ↔ NiEu + fcc_A1	810	Eutectic	0.997	XRD/DTA/metallography	[38, 46]
	810		0.997	Calculated	[13]
	750		0.954	Calculated	This work

Comparison between the experimental data (*T*/°C), *x*(Liq,Eu), *x*(Liq1,Eu), *x*(Liq2,Eu)) and the calculated values after the optimizations using the Redlich–Kister equation [14]

formation of the intermetallic compounds of the Ni–Yb system were compared, and the enthalpies and entropies calculated in the present work are added; it is noted that the values of the enthalpies are in general in good agreement with the measured, predicted or calculated ones by Berezutskii et al. [13], but the optimized entropies of formation in the present work are higher than the ones calculated by Berezutskii et al. [13]. In Table 4, the calculated enthalpies and entropies of formation of the intermetallic compounds for the four systems Ni–Yb, Ni–Ho, Ni–Eu and Ni–Sc are compared. We noted that the optimized entropies of formation of the compounds having the same chemical formula have the same order of magnitude and are higher than those published in [13]. The calculated phase diagram of the Ni–Yb system compared to the experimental results of Berezutskii et al. [13] is shown in Fig. 3. A good agreement is noted between the calculated phase diagram and the experimental data. The calculated invariant reactions are compared with the experimental ones in Table 5. A satisfactory agreement is noted.

The calculated integral mixing enthalpies of the liquid phase compared to those of [13] are shown in Fig. 4. We noted that the integral mixing enthalpies of liquid phase are independent of temperature. Hence, the curves plotted for $T = 1000$, 1500 and 1800 K are superimposed. The calculated free Gibbs energies of mixing of liquid at 1000, 1500 and 1800 K are shown in Fig. 5. The calculated activities of Ni in Yb and Yb in Ni at $T = 1600$ K are shown in Fig. 6. A good agreement was observed with the experimental data measured by Berezutskii et al. [13]. Within the experimental uncertainties, most of the experimental data are well reproduced by the present calculation. The calculated enthalpies of formation of the intermetallic compounds listed in Table 2 are shown in Fig. 7. A good agreement is noted with the various experimental data [13, 22, 26, 27] and the predicted ones [25].

Ni–Eu system

In Table 6 the optimized thermodynamic parameters during this work are listed. The Eu–Ni phase diagram calculated from our optimized database is shown in Fig. 8. There are some discrepancies with the diagram published by Berezutskii et al. [13] especially the immiscibility range ($0.1 < x_{Ni} < 0.4$ at.) at 1600 K. The calculated enthalpies of mixing of the liquid phase at 1600 K are shown in Fig. 9. The calculated enthalpies and entropies of formation of the intermetallic compounds are shown in Fig. 10 and listed in Table 7. The invariant reactions are compared with the experimental ones in Table 8.

A satisfactory agreement exists between the calculated temperatures of these invariant reactions in the present work and the data by Berezutskii et al. [13], except for the $Liq + Ni_5Eu \leftrightarrow Ni_{17}Eu_2$ reaction, with some data from the literature. A good agreement is noted with the data by Berezutskii et al. [13], but as for the Ni–Yb system a noticeable discrepancy exists for the entropies.

Conclusions

The Ni–Yb and Ni–Eu binary systems were assessed by the CALPHAD method and the Redlich–Kister formalism for the solution phases. The two experimental phase diagrams and a calculated one reported by different authors for each system are discussed. The optimization of a set of coherent thermodynamic parameters was obtained for both systems by considering the most probable phase diagram and the available thermodynamic data. A good agreement was obtained for both systems between the experimental and calculated data.

References

1. Buschow KHJ. Intermetallic compounds of rare-earth and 3d transition metals. Rep Prog Phys. 1977;40:1179–256.
2. Schlapbach L, Züttel A. Hydrogen-storage materials for mobile applications. Nature. 2001;414:353–8.
3. Pecharsky VK, Gschneidner KA. Giant magnetocaloric effect in $Gd_5(Si_2Ge_2)$. Phys Rev Lett. 1997;78:4494–7.
4. Quigg RJ. New alloy developments in single crystal and DS alloys. High Temp Mater Process (Lond). 1993;11(1–4):247–54.
5. Narita T, Thosin KZ, Fengqun L, Hayashi S, Murakami H, Gleeson B, Young D. Development of Re-based diffusion barrier coatings on nickel based superalloys. Mater Corros. 2005;56(12):923–9.
6. Lee K, Heiman N. Magnetism in rare earth-transition metal amorphous alloy films. In: Magnetism and magnetic materials: 20th annual conference, vol 24, no 1. San Francisco: AIP Publishing; 1974, p 108–9.
7. Louzguine DV, Yavari AR, Inoue A. Mischmetal as an alloying addition to amorphous materials and glass formers. J Non-Cryst Solids. 2003;316(2):255–60.
8. Spassov T, Lyubenova L, Köster U, Baró MD. Mg–Ni–RE nanocrystalline alloys for hydrogen storage. Mater Sci Eng. 2004;A375–A377:794–9.
9. Maezawa R, Kashimoto S, Ishimasa T. Icosahedral quasicrystals in Zn–T–Sc (T = Mn, Fe, Co or Ni) alloys. Philos Mag Lett. 2004;84(4):215–23.
10. Kaufman L, Bernstein H. Computer calculation of phase diagram. New York: Academic Press; 1970.
11. Kardellass S, Servant C, Selhaoui N, Iddaoudi A, Amar MA, Bouirden L. Thermodynamic description of the Ni–Sc system. Calphad. 2013;42:59–65.
12. Kardellass S, Servant C, Selhaoui N. Thermodynamic description of the holmium-nickel system. J Therm Anal Calorim. 2015;119–2:1225–34.
13. Berezutskii VV, Shevchenko MA, Ivanov MI, Sudavtsova VS. Thermodynamic properties of liquid alloys Ni–Eu and Ni–Yb. Russ J Phys Chem A. 2014;88–9:1463–71.
14. Redlich O, Kister AT. Algebraic representation of thermodynamic properties and the classification of solutions. Ind Eng Chem. 1948;40(2):345–48.

15. Buschow KHJ. Note on the structure and occurrence of ytterbium transition metal compounds. J Less Common Met. 1972;26:329–33.
16. Palenzona A, Cirafici S. The ytterbium–nickel system. J Less Common Met. 1973;33:361–7.
17. Nash P. The nickel–ytterbium system. Bull Alloy Phase Diagr. 1989;10(2):129–32.
18. Gschneidner KA Jr, Calderwood FW. Intra rare earth binary alloys: phase relationships, lattice parameters and systematics. In: Gschneidner Jr KA, Eyring L, editors. Handbook on the physics and chemistry of rare earths, vol. 8. Amsterdam: North-Holland Physics Publishing; 1986. p. 1–161.
19. Massalski B. Binary alloy phase diagrams, vol. 1, 2. Metals Park: American Society for Metals; 1986.
20. Okamoto H. Ni–Tb (nickel–terbium). J Phase Equilib Diffus. 2005;26-3:295–7.
21. Ensslen K, Bucher E, Oesterreicher H. Hydrides and valance changes in some compounds of Yb–Ni, Yb–Pd and related systems. J Less Common Met. 1983;92:343–53.
22. Bayanov AP. Enthalpies of formation of compounds of the rare earth elements, calculated on the basis of their crystal chemistry. Izv Akad Nauk SSSR Neorg Mater. 1973;9(6):959–63 **(in Russian; TR: Inorg. Mater. 1973; 9(6):858–861)**.
23. Kubaschewski O. Physical chemistry of metallic solutions and intermetallic compounds N.P.L. Syrup. No. 9, London; 1959.
24. Gschneidner KA Jr. Valences of europium and ytterbium in compounds. J Less Common Met. 1969;17:13–24.
25. De Boer MFR, Boom R, Mattens WCM, Miedema AR. Cohesion in metals, transition metals alloys. Amsterdam: North-Holland; 1988.
26. Brutti S, Ciccioli A, Balducci G, Gigli G, Borzone G, Raggio R, Ferro R. Thermodynamics of the Ni–Yb system. J Phase Equilib. 2002;23:1.
27. Di Pascasio F, Gozzi D, Parodi N, Borzone G. Thermodynamic properties of Me_2Ni_{17} (Me = Sm, Dy, and Yb) intermetallics by solid electrolyte cells under effusion conditions. J Phys Chem B. 2002;106–16:4285–93.
28. Cignini PL, Gozzi D, Iervolino M, Latini A. Thermodynamics of intermetallics through emf measurements under effusion conditions. Intermetallics. 2003;11–12:1167–74.
29. Gozzi D, Iervolino M. Thermodynamics of Ni richest intermetallics along the rare-earth series. Intermetallics. 2005;13(11): 1172–83.
30. Rojas DP, Barquín LF, Echevarria-Bonet C, Fernández JR. $YbNi_2$: a heavy fermion ferromagnet. Solid State Commun. 2012;152(19):1834–7.
31. Nesbitt EA, Williams HJ, Wernick JH, Sherwood RC. Magnetic moments of intermetallic compounds of transition and rare-earth elements. J Appl Phys. 1962;33(5):1674–8.
32. Bleaney B. Magnetic moments of the lanthanon–nickel ($LnNi_5$) compounds. Proc Phys Soc. 1963;82:469–71.
33. Nowik I, Ofer S, Wernick JH. Relaxation phenomena in mossbauer spectra of ^{170}Yb in intermetallic compounds of ytterbium. Phys Lett. 1967;24(2):89–90.
34. Carfagna PD, Wallace WE. Magnetic characteristics of 2–17 lanthanide–nickel compounds. J Appl Phys. 1968;39(11):5259–62.
35. Nowik I, Dunlap BD. Crystalline fields and exchange interactions in $YbNi_2$. J Phys Chem Solids. 1973;34:465–71.
36. Klaasse JCP, Martens WCM, De Boer ER, De Chatel PF. Magnetic properties of ytterbium intermetallic compounds with intermediate valency. Physica. 1977;86-88B:234–6.
37. Bleaney B. Crystal field effects and the cooperative state II. The cubic $LnNi_2$ compounds. Proc R Soc A. 1963;276:28–38.
38. Savitskii EM, Terekhova VF, Torchinova RS. Europium alloys. Metalloved Term Obrab Met. 1967;2–25; **TR: Met. Sci. Heat Treat. 1967; 1/2:100–106**.
39. Chechernikov VI, Speranskiy NM, Terekhova VF, Rozhkova RS. Some magnetic properties of Ni–Eu alloys. Fiz Met Metalloved. 1965;20(1):157–9 **(in Russian;TR: Phys. Met. Metallogr. (USSR). 1965; 20(1):146–147)**.
40. Terekhova VE, Kripyakevich PI, Frankevich DP, Torchinova RS. Compounds of Eu_2Ni_{17} and $EuNi_5$ and their crystal structure. Izv Akad Nauk SSSR Met. 1967;1:205–20 **(in Russian)**.
41. Kripyakevich PI, Gladlshevs'kii EI. The compounds $EuNi_6$, $TmNi_6$, and $LuNi_6$ and their crystal structure. Izv Akad Nauk Ukr RSR. 1966;6:769–71 **(in Ukrainian)**.
42. Palenzona A. New phases of MXs-type compounds of Eu and Yb with a $CaCu_5$ type structure. Cl Sci Fis Mat Nat. 1967; 42(4):504–9 **(in Italian)**.
43. Gavra Z, Murray JJ, Calvert LD, Taylor JB. The $EuNi_5$–H system. J Less Common Met. 1985;105(2):291–301.
44. Moriarty JL, Humphreys JE, Gordon RO, Baenziger NC. X-ray examination of some rare earth containing binary alloy systems. Acta Crystallogr. 1966;21:840–1.
45. Olcese GL. Crystal structure and magnetic properties of some 7:3 binary phases between lanthanides and metals of the 8th group. J Less Common Met. 1973;33:71–81.
46. Tung CH, Nash P. The Eu–Ni (europium–nickel) system. Bull Alloy Phase Diagr. 1989;10(2):127–9.
47. Ning YT, Zhou XM, Zhen Y,Chen NY, Xu H, Zhu JZ. The prediction and synthesis of some new intermetallic compounds between transition metals and rare earth metals. J Less Common Met. 1989;147(2):167–73.
48. Okamoto H. Eu–Ni (europium–nickel). J Phase Equilib. 1992;13(4):441–2.
49. Oliver EW, West KW, Cohen RL, Buschow KHJ. Mössbauer effect of Eu in $EuNi_5$, $EuMg_2$ and their hydrides. J Phys F Met Phys. 1978;8(4):701–7.
50. Dinsdale AT. SGTE data for pure elements. CALPHAD. 1991;15–4:317–425.
51. Inden G. Approximate description of the configurational specific heat during a magnetic order-disorder transformation. In: Proceeding of the 5th project meeting CALPHAD. Max Planck Institute for Iron Research. Düsseldorf: Max-Planck-Institut für Eisenforschung GmbH; 1976. p. III-4.1–III-4.13.
52. Inden G. The role of magnetism in the calculation of phase diagrams. Phys B + C. 1981;103-1:82–100.
53. Hillert M, Jarl M. A model for alloying effects in ferromagnetic metals. Calphad. 1987;2:227–38.
54. Grimvall G. Thermophysical properties of materials. Amsterdam: Elsevier; 1999.
55. Jansson B. Ph. D. thesis, Royal Institute of Technology, Stockholm, Sweden; 1984
56. Sundman B, Jansson B, Andersson JO. The thermodynamic databank system. Calphad. 1985;2:153–90.
57. Lupis CH. Chemical thermodynamics of metals. North-Holland: Prentice-Hall Inc.; 1983.
58. Kumar KC, Wollants P. Some guidelines for thermodynamic optimization of phase diagrams. J Alloy Compd. 2001; 320–2:189–98.
59. Haszko SE. Intermediate phases with the Cu_5Ca structure. Trans AIME. 1960;218:763.
60. Lemaire R, Paccard D. Crystal structures of intermetallic compounds between nickel and rare earth metals of yttrium. Bull Soc Fr Miner Crist. 1969;92:9–16.
61. Rosa JW, Walley SP. Conf Ser Inst Phys. 1978;37:155
62. Haszko SE. Intermediate phases with the $MgCu_2$ structure. Trans AIME. 1960;218:958.